

Pen and Paper Cryptography: Encryption and Decryption

Introduction

Cryptography is used to convert readable information into a form that cannot be easily understood by an unauthorized person. The original readable message is called plaintext, while the encrypted message is called ciphertext. Encryption converts plaintext into ciphertext, and decryption converts it back into the original message.

In this exercise, three simple substitution techniques are used: Caesar Cipher, Shift Cipher, and ROT13. The calculations are performed using the alphabet positions, while spaces are kept unchanged.

Alphabet Used

For the calculations, the English alphabet is considered as:

ABCDEFGHIJKLMNOPQRSTUVWXYZ

The alphabet positions are:

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

1. Encryption

a. Caesar Cipher

For the Caesar Cipher, a shift of 3 is used.

The rule is:

$$C = (P + 3) \bmod 26$$

where P is the plaintext letter and C is the encrypted letter.

Plaintext:

CHASE YOUR DREAMS

Applying a shift of 3:

C	H	A	S	E
F	K	D	V	H

Therefore,

CHASE YOUR DREAMS

becomes

FKDVH BRXU GUHDPV

Encrypted Text:

FKDVH BRXU GUHDPV

b. Shift Cipher with Key 7

For a shift cipher with key 7, every letter is shifted seven positions forward.

The encryption rule is:

$$C = (P + 7) \bmod 26$$

Plaintext:

CHASE YOUR DREAMS

Some letter conversions are:

C	H	A	S	E	Y	O
J	O	H	Z	L	F	V

Continuing the same shift for all letters gives:

CHASE YOUR DREAMS

|

| Shift by 7

v

JOHZL FVBY Y YLHTZ

Therefore, the encrypted message is:

JOHZL FVBY KYLHTZ

c. ROT13

ROT13 is a special substitution cipher where every letter is rotated by 13 positions. Applying ROT13 twice gives the original message.

The rule is:

$$C = (P + 13) \bmod 26$$

Plaintext:

CHASE YOUR DREAMS

Some conversions are:

C	H	A	S	E	Y
P	U	N	F	R	L

Therefore:

CHASE YOUR DREAMS

becomes:

PUNFR LBHE QERNZF

Encrypted Text:

PUNFR LBHE QERNZF

2. Decryption

a. Caesar Cipher

Ciphertext:

WRWDOOB VHFUHW PHVVDJH

For decryption using a Caesar Cipher with shift 3, each letter is shifted three positions backward.

The decryption rule is:

$$P = (C - 3) \bmod 26$$

For example:

$$W \rightarrow T \quad R \rightarrow O \quad W \rightarrow T$$

Similarly,

$$VHFUHW \rightarrow SECRET$$

and

$$PHVVDJH \rightarrow MESSAGE$$

Therefore:

WRWDOOB VHFUHW PHVVDJH

decrypts to:

TOTALLY SECRET MESSAGE

Plaintext:

TOTALLY SECRET MESSAGE

b. Shift Cipher with Key 5

Ciphertext:

SJI IBTW YNRJ

For decryption, each letter is shifted five positions backward.

The rule is:

$$P = (C - 5) \bmod 26$$

For example:

$$S \rightarrow N$$

$$J \rightarrow E$$

$$I \rightarrow D$$

Therefore:

$$SJI \rightarrow NED$$

Similarly:

$$IBTW \rightarrow DWOR$$

and

$$YNRJ \rightarrow TIME$$

Thus, using the ciphertext exactly as given in the question, the decrypted text is:

NED DWOR TIME

Note: The result does not form a meaningful English sentence. This suggests that there may be a typing error in the given ciphertext. The decryption method used here is the standard backward shift of 5.

c. ROT13

Ciphertext:

FRAQ ZBER PBSSRR

ROT13 is its own inverse, so applying ROT13 again decrypts the message.

Some conversions are:

$$F \rightarrow S \quad R \rightarrow E \quad A \rightarrow N \quad Q \rightarrow D$$

Therefore:

FRAQ ZBER PBSSRR

|

| ROT13

v

SEND MORE COFFEE

Hence the decrypted message is:

SEND MORE COFFEE

Summary

Method	Operation	Result
Caesar Cipher	Encrypt “CHASE YOUR DREAMS” using shift 3	FKDVH BRXU GUHDPV
Shift Cipher (7)	Encrypt “CHASE YOUR DREAMS”	JOHZL FVBY KYLHTZ
ROT13	Encrypt “CHASE YOUR DREAMS”	PUNFR LBHE QERNZF
Caesar Cipher	Decrypt “WRWDOOB VH-FUHW PHVVDJH”	TOTALLY SECRET MESSAGE
Shift Cipher (5)	Decrypt “SJI IBTW YNRJ”	NED DWOR TIME
ROT13	Decrypt “FRAQ ZBER PB-SSRR”	SEND MORE COFFEE

Conclusion

The exercise demonstrates how simple substitution ciphers can be used to encrypt and decrypt messages. The Caesar Cipher and Shift Cipher use a fixed number of positions to move each letter in the alphabet, while ROT13 always uses a shift of 13. These methods are easy to perform using pen and paper and are useful for understanding the basic idea of encryption and decryption.

However, these ciphers are not secure enough for protecting real-world sensitive information because the possible number of keys is very small. Modern cryptographic systems use much stronger algorithms and larger key spaces.